R7.3: conductor-sensitive algorithms across an ℓ-volcano gap

Second toy test of Conjecture 10′. Native Rust, $p \in [2^{20}, 2^{22})$, counted operations only. Source: `examples/r7_3_conductor_sensitive.rs`; raw data `raw/`; full text `RESULTS.md`.

Hypothesis H. Some algorithm whose cost depends on $\mathrm{End}(E)$ shows a measurable crater-vs-floor cost gap on ℓ-volcanoes.
Falsifier. Every crater/floor ratio CI contains 1 after the automorphism baseline is removed (null), or a CI excludes 1 (report size, cause, and whether it exceeds the bounded factor).

Verdict (measured)

1. **Iterations and relation yield: null.** Conductor effect after removing $|\mathrm{Aut}|$: 1.0018 [0.9986, 1.0050] at $j = 0$ and 0.9997 [0.9966, 1.0030] at $j = 1728$ (32 instances each); endomorphism-mixed rho, iterations 1.0001 [0.9921, 1.0080] (40 instances); plain rho 0.9999 [0.9939, 1.0061]; class factor at $h(D) = 2, 3$ craters 0.99985 [0.9993, 1.0004] (56 instances); transported-base census identical to its source in 120/120 checks. Conjecture 10′ is untested-but-not-contradicted at this scale for these algorithms.
2. **Counted multiplications: the falsifier's second branch fires for one algorithm.** The endomorphism-mixed walk costs the floor $1.46 \times$ [1.44, 1.48] the crater, with equal iteration counts; the gap is the cost of evaluating the cheapest explicit endomorphism ($2.72 \times$ [2.59, 2.85]: three Vélu maps on the floor, one on the crater; 2.5 to 2.9 $\times$ for $\ell = 5$ to 13). That walk is $1.22 \times$ (crater) and $1.79 \times$ (floor) dearer than plain rho and takes no fewer iterations (0.9993 [0.989, 1.009]), so the best strategy ignores ω and has crater/floor cost 0.9999 [0.9945, 1.0055]. A polynomial-in-ℓ constant on an inferior algorithm, not a change in the best attack's cost.

The known automorphism effect is the positive control: ζ-fold rho is cheaper than ± 1 rho on the crater by 0.5534 [0.5511, 0.5559] ($j = 0$, ideal 0.5774) and 0.6820 [0.6791, 0.6848] ($j = 1728$, ideal 0.7071).

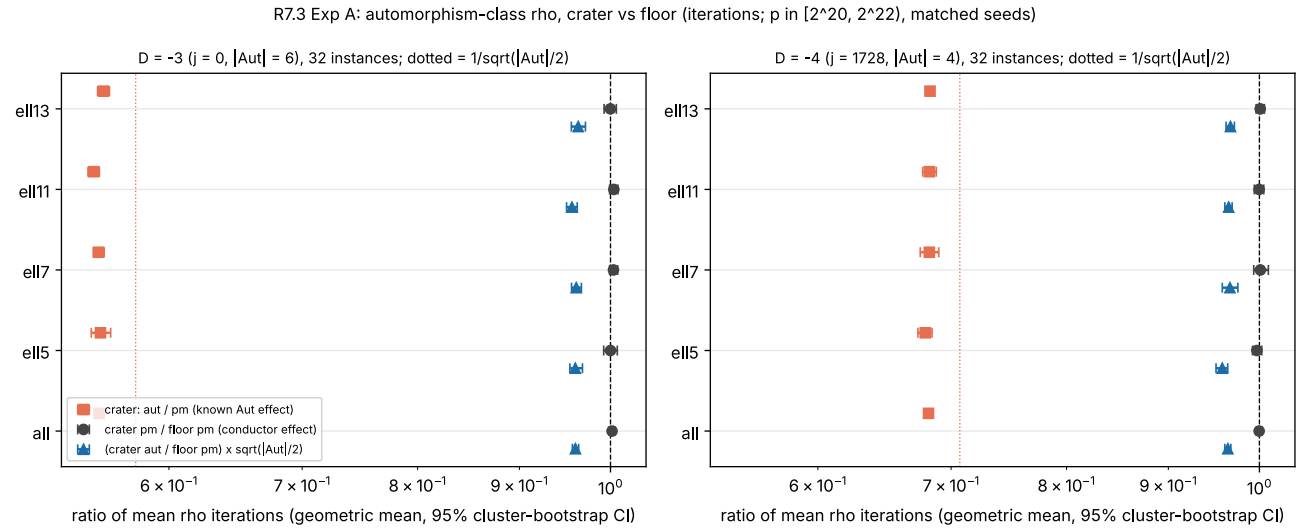


Figure 1: Exp A, 32 instances per D, 3000 runs per curve and variant. Squares: aut/pm on the crater (known Aut effect; dotted line is the ideal $\frac{1}{\sqrt{|\mathrm{Aut}|/2}}$). Circles: crater pm / floor pm, the conductor effect with the baseline removed. Triangles: (crater aut / floor pm) times the ideal baseline factor; the offset from 1 is the implementation shortfall of the 6- and 4-fold classes, not a conductor effect. Geometric means, 95% cluster-bootstrap CIs.

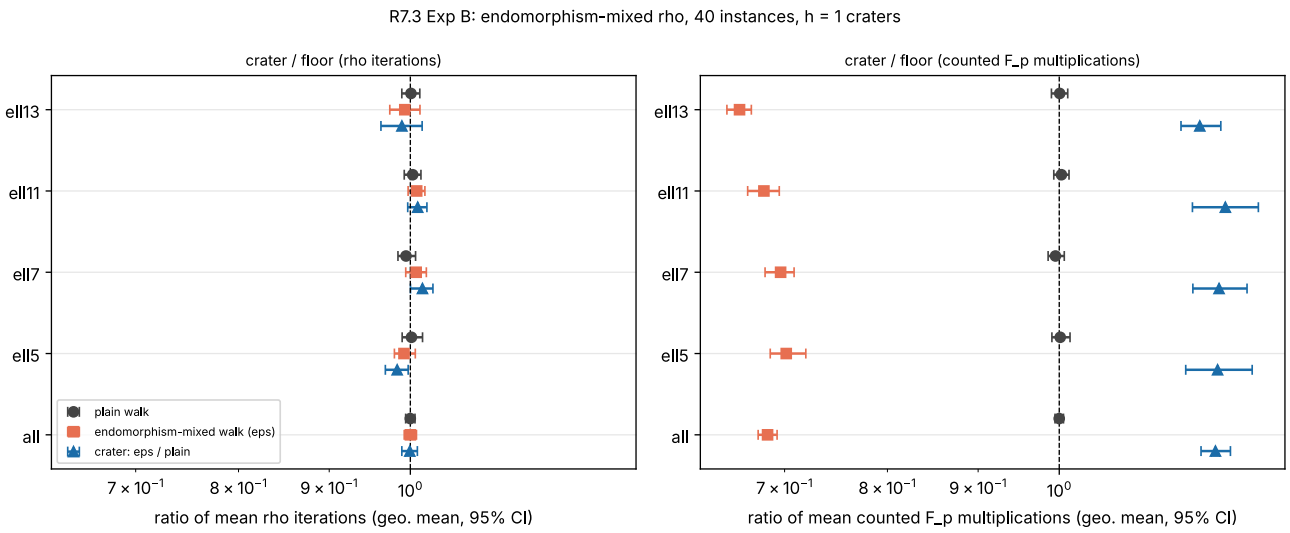


Figure 2: Exp B, 40 instances ($h = 1$ craters, 380 floors), 600 runs per curve and variant. Left: iterations; right: counted F_p multiplications. Crater/floor ratios for the plain walk and the endomorphism-mixed walk (2 of 16 table slots), and eps/plain on the crater.

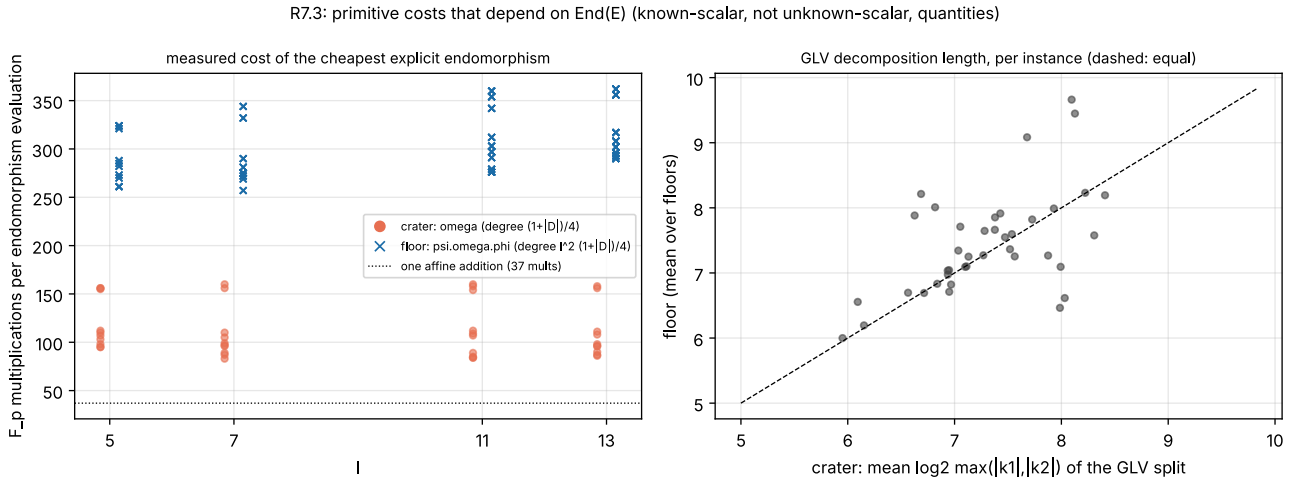


Figure 3: Left: measured multiplications per endomorphism evaluation (crater ω; floor $\psi\omega\varphi$, degree $\ell^2 \frac{1+|D|}{4}$), one dot per curve; an affine addition costs 32 to 42. Right: GLV split length $\log_2 \max(|k_1|, |k_2|)$ of the measured eigenvalue, floor against crater per instance (floor minus crater: $+0.15$ bits, sd 0.69, n = 40).

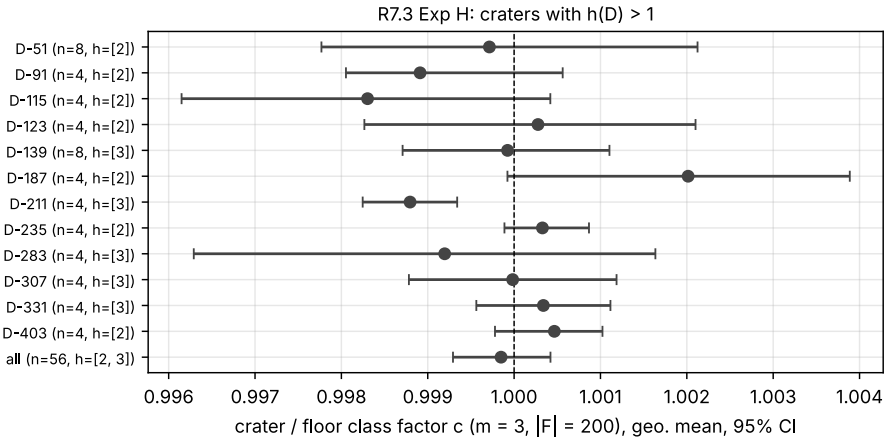


Figure 4: Exp H, 56 instances, $h(D) \in \{2, 3\}$, 12 discriminants, $m = 3$, $|F| = 200$: crater over mean of its floors, class factor c. 5 of 48 per-discriminant intervals exclude 1 (by at most 0.34%, both signs; unadjusted).

Methods in brief

Instances: $p = \frac{t^2+|D|v^2}{4}$, $\ell \parallel v$, $t \not\equiv 2 (\mathrm{mod}\, \ell)$; crater = CM curve of D (End $= \mathcal{O}_K$), floors = other F_p-roots of $\Phi_{\ell(j_0,Y)}$; $\#E = N$ certified on every curve; the DLP lives in the prime-order subgroup, on which every vertical isogeny is an isomorphism. Walk: 16-entry r-adding walk on orbit classes, exact collision bookkeeping, every recovered k verified (1 608 000 runs, 0 wrong, 0 failed). Endomorphisms: explicit Vélu maps, certified on random points (on-curve and additive) with eigenvalues matched (ω from the π-eigenline of $E[n_\omega]$; floor $\psi\omega\varphi$ acts as $\pm\ell\lambda$). Seeds: the same step-table and start scalars and hidden k on every curve of an instance.

Limits

Toy sizes, one vertical step, $\ell \leq 13$, $h \leq 3$; endomorphism algorithms only for $h = 1$ ($D \in \{-19, -43, -67, -163\}$); endomorphism slot fraction fixed at 1/8; the library `LineOracle` / F_{p^2} ζ-fold and a level-dependent index-calculus oracle were not run; the iteration counts are counts, not timings. No sub-rho claim, no claim about cryptographic sizes or any deployed curve.